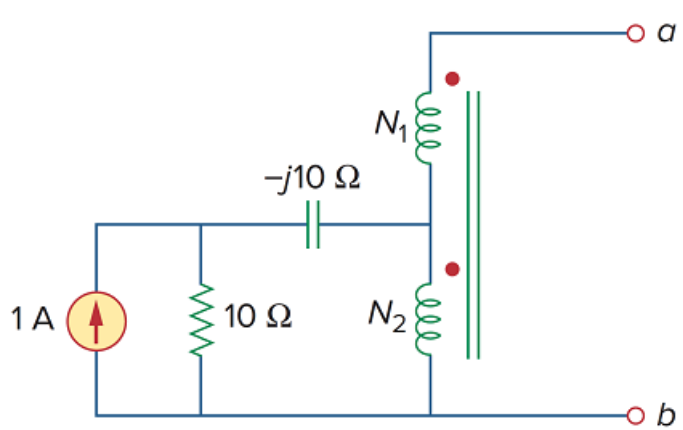


Problem

In the circuit of Fig. 13.131, $N_1 = 190$ turns and $N_2 = 10$ turns. Determine the Thevenin equivalent circuit looking into terminals a and b . What would be the value of Z_L that would absorb maximum power from the circuit?

Figure 13.132



Step-by-step solution

Step 1 of 13

Refer to Figure 13.132 from the textbook.

Observe that the circuit is a step-up auto transformer.

Use the Current to Voltage source transformation and replace the current source by an equivalent voltage source with 10Ω resistor in series with it.

Calculate the equivalent voltage source, V .

$$V = IR$$

Substitute 1 A for I and 10Ω for R .

$$\begin{aligned} V &= (1)(10) \\ &= 10 \text{ V} \end{aligned}$$

Step 2 of 13

Sketch the modified circuit as shown in Figure 1.

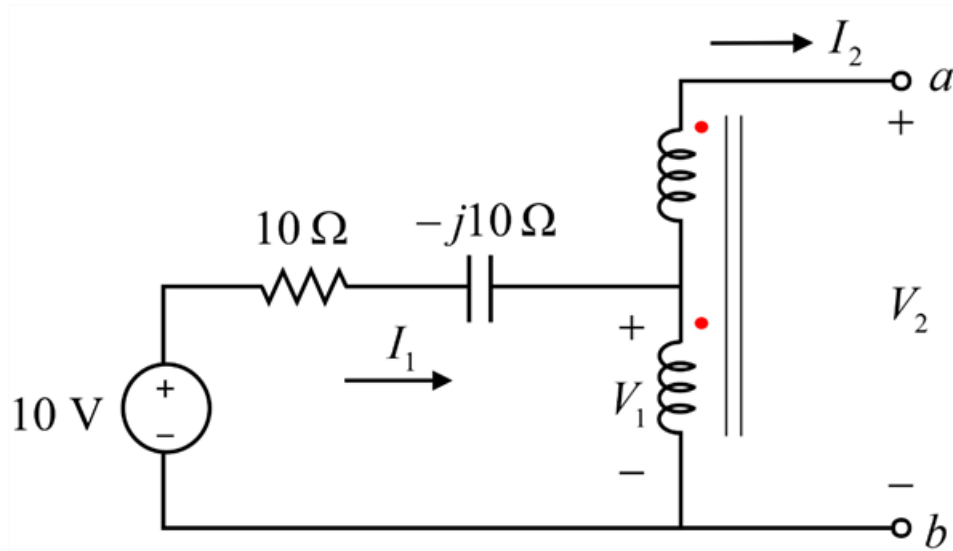


Figure 1

Step 3 of 13

For the step-up transformer,

$$\frac{V_1}{V_2} = \frac{N_2}{N_1 + N_2} \dots\dots (1)$$

Here,

V_1 is the voltage on the primary side of the transformer,

V_2 is the voltage on the secondary side of the transformer.

Step 4 of 13

Substitute 190 for N_1 and 10 for N_2 in equation (1).

$$\frac{V_1}{V_2} = \frac{10}{190 + 10}$$

$$\frac{V_1}{V_2} = \frac{10}{200}$$

$$\frac{V_1}{V_2} = \frac{1}{20}$$

$$\frac{V_1}{V_2} = 0.05 \dots\dots (2)$$

Substitute 10 V for V_1 .

$$\frac{10}{V_2} = 0.05$$

$$V_2 = \frac{10}{0.05}$$

$$V_2 = 200 \text{ V}$$

Step 5 of 13

The open circuit voltage across the terminals a and b or the Thevenin Voltage (V_{th}) is the same as V_2 .

Thus,

$$\begin{aligned} V_{th} &= V_2 \\ &= 200 \text{ V} \end{aligned}$$

Step 6 of 13

Calculate the Thevenin impedance, Z_{th} .

Replace the voltage source by a short-circuit and connect a 1 V voltage source across a and b as shown in Figure 2.

Thus,

$$Z_{th} = \frac{1}{I_2} \dots\dots (3)$$

Step 7 of 13

Sketch the modified circuit as shown in Figure 2.

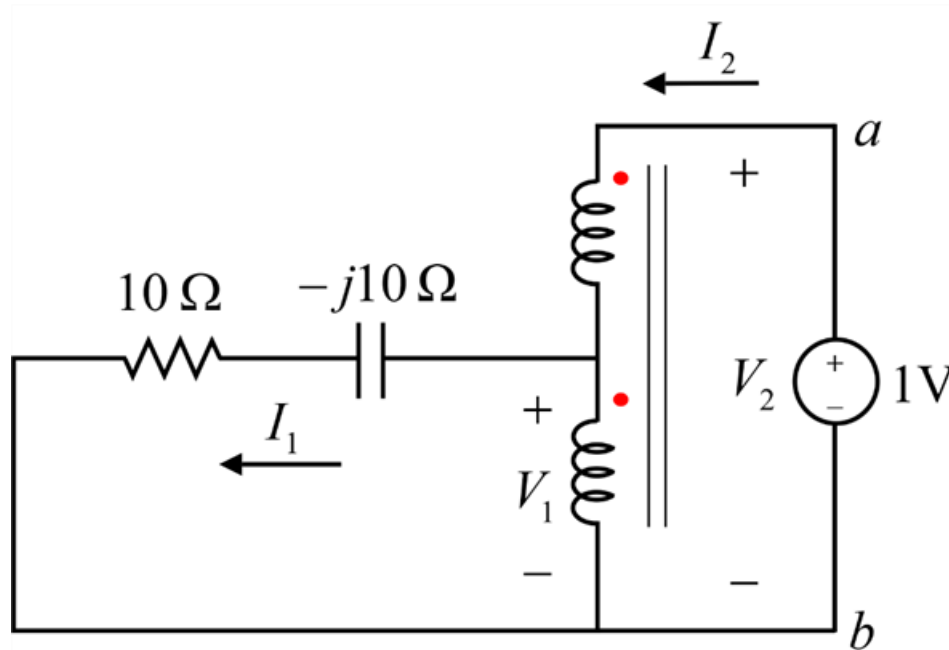


Figure 2

Step 8 of 13

From equation (2),

$$\begin{aligned} \frac{V_1}{V_2} &= 0.05 \\ V_1 &= 0.05V_2 \end{aligned}$$

Substitute 1 V for V_2 .

$$\begin{aligned} V_1 &= (0.05)(1) \\ &= 0.05 \text{ V} \end{aligned}$$

Step 9 of 13

Since mmf is balanced in a transformer.

Thus,

$$I_2 N_1 = (I_1 - I_2) N_2$$

$$(N_1 + N_2) I_2 = I_1 N_2$$

$$I_1 = \left(\frac{N_1 + N_2}{N_2} \right) I_2$$

Substitute 190 for N_1 and 10 for N_2 .

$$I_1 = \left(\frac{190 + 10}{10} \right) I_2$$

$$= 20 I_2$$

Step 10 of 13

Apply Kirchhoff's Voltage law on the primary side of the autotransformer circuit shown in Figure 2.

$$0 = 10 I_1 - j10 I_1 - V_1$$

$$V_1 = (10 - j10) I_1$$

Substitute $20 I_2$ for I_1 and 0.05 V for V_1 .

$$0.05 = (10 - j10)(20 I_2)$$

$$I_2 = \frac{0.05}{20(10 - j10)}$$

$$I_2 = 1.25 \times 10^{-4} + j1.25 \times 10^{-4} \text{ A}$$

Step 11 of 13

Substitute $1.25 \times 10^{-4} + j1.25 \times 10^{-4} \text{ A}$ for I_2 in equation (3).

$$Z_{th} = \frac{1}{1.25 \times 10^{-4} + j1.25 \times 10^{-4}}$$

$$= 4000 - j4000 \Omega$$

$$= 4 - j4 \text{ k}\Omega$$

Step 12 of 13

Sketch the Thevenin equivalent circuit as shown in Figure 3.

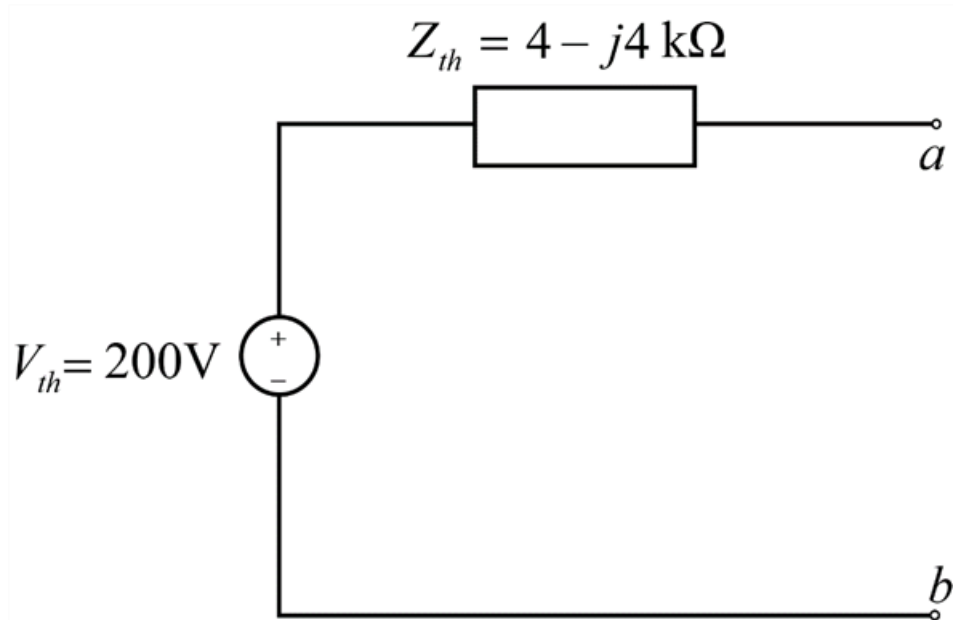


Figure 3

Step 13 of 13

Use maximum power transfer theorem to determine the load impedance Z_L that will absorb maximum power from the circuit.

Thus,

$$Z_L = Z_{th}^*$$

Substitute $4 - j4 \text{ k}\Omega$ for Z_{th} .

$$\begin{aligned} Z_L &= (4 - j4)^* \\ &= 4 + j4 \text{ k}\Omega \end{aligned}$$

Therefore, the Thevenin voltage, V_{th} is $\boxed{200 \text{ V}}$, the Thevenin impedance, Z_{th} is $\boxed{4 - j4 \text{ k}\Omega}$ and the load impedance Z_L that will absorb maximum power from the circuit is $\boxed{4 + j4 \text{ k}\Omega}$.